

Credit Risk Analysis using Regression

Gabriel ISSA

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Package set up

```
#install.packages("stringr")
#install.packages("stargazer")
#install.packages("ggplot2")
#install.packages("dplyr")
#install.packages("tidyr")
#install.packages("corrplot")
#library(stargazer)
library(stringr)
library(ggplot2)
library(dplyr)
library(tidyr)
library(corrplot)
```

Probability models conception

Model design

The first step we need to do is to import the data and decide which variables we want to include in the model

```
raw_df = read.csv("raw_data/credit-1.csv", stringsAsFactors = TRUE)
summary(raw_df)
```

```
##           id           default      seniority           home
##  Min.      : 1      Min.      :0.0000      Min.      : 0.000      ignore : 20
## 1st Qu.:1112      1st Qu.:0.0000      1st Qu.: 2.000      other  : 319
## Median :2224      Median :0.0000      Median : 5.000      owner  :2106
## Mean   :2224      Mean   :0.2809      Mean    : 7.991      parents: 782
## 3rd Qu.:3335      3rd Qu.:1.0000      3rd Qu.:12.000      priv   : 246
## Max.    :4446      Max.    :1.0000      Max.    :48.000      rent   : 973
##          age          marital          job          expenses
##  Min.    :18.00      divorced : 38      fixed    :2803      Min.    : 35.0
## 1st Qu.:28.00      married  :3238      freelance:1021      1st Qu.: 35.0
## Median :36.00      separated: 130      others   : 171      Median : 51.0
## Mean    :37.08      single   : 973      partime  : 451      Mean    : 55.6
## 3rd Qu.:45.00      widow    : 67              3rd Qu.: 72.0
## Max.    :68.00              Max.    :180.0
```

```
##      income      assets      debt      amount
## Min.   : 1.0   Min.   : 0   Min.   : 0.0   Min.   : 100
## 1st Qu.: 90.0  1st Qu.: 0   1st Qu.: 0.0   1st Qu.: 700
## Median :124.0  Median : 3000  Median : 0.0   Median :1000
## Mean   :140.6  Mean   : 5355  Mean   : 342.3  Mean   :1039
## 3rd Qu.:170.0  3rd Qu.: 6000  3rd Qu.: 0.0   3rd Qu.:1300
## Max.   :959.0  Max.   :300000  Max.   :30000.0  Max.   :5000
##      savings
## Min.   : -8.160
## 1st Qu.: 1.615
## Median : 3.120
## Mean   : 3.860
## 3rd Qu.: 5.196
## Max.   :33.250
```

Dependent variable :

- *default* : 1 if the client made a default 0 else

Raw variables to include :

- *senority* : (negative) Longer job tenure signals stable income and lower risk of sudden job loss
- *age* : (negative) older borrowers are more experienced when managing their money and often have more stable financial situation which lead to lower default risk

Variables to transform :

- *Is_house_owner* is equal to 1 if the client own a house
 - Intuition : (negative) Homeowners have a valuable asset and may have already loan money to buy it so I expect lower default.
- *is_married* is equal to 1 if the client is married else 0
 - Intuition : (negative) married couple have to income to reimburse a debt and a more stable situation which imply lower default risk.
- *has_fixed_job* : 1 if job = “fixed” else 0
 - Intuition : (negative) a full time salary job mean stable income and so less financial situation volatility
- *is_freelance* : 1 if job = “freelance” else 0
 - Intuition : (positive) a freelance job is synonym of volatile income and changing financial situation which induce higher default risk
- *is_parttime* : part-time job are often mean limited and unstable income
- *reimbursement_cap* : $\text{expenses} / \text{income}$
 - Intuition : (positive) High expenses leave little disposable income to service the loan, so the expenses-to-income ratio should be positively related to default.
- *income_to_total_debt* : $\text{income} / (\text{debt} + \text{amount})$
 - Intuition : (negative) the higher is the debt compared to income the hardest it will be for the client to reimburse it

- *savings_to_total_debt* : $\text{savings} / (\text{debt} + \text{amount})$
 - Intuition : (negative) Savings are a liquidity cushion that lets a borrower keep paying during a temporary income shock, so the lower it is compared to debt the higher the risk
- $\log(1+\text{income})$:
 - Intuition : (negative) Higher income means more capacity to absorb repayments and unexpected shocks, so I expect a negative sign. I use the log because the effect shows diminishing returns.
- $\log(1+\text{asset})$:
 - Intuition : (negative) same principle than income

To exclude of the analysis :

- *id* : identifier of the client

Final regression equation :

This corresponds to the following final equation :

$$\begin{aligned} \text{default}_i = & \beta_0 + \beta_1 \text{seniority}_i + \beta_2 \text{age}_i \\ & + \beta_3 \text{Is_house_owner}_i + \beta_4 \text{Is_married}_i \\ & + \beta_5 \text{has_fixed_job}_i + \beta_6 \text{is_freelance}_i + \beta_7 \text{is_parttime}_i \\ & + \beta_8 \text{reimbursement_cap}_i + \beta_9 \text{income_to_total_debt}_i \\ & + \beta_{10} \text{savings_to_total_debt}_i \\ & + \beta_{11} \ln(1 + \text{income}_i) + \beta_{12} \ln(1 + \text{assets}_i) + \varepsilon_i \end{aligned}$$

Data Preparation

The data needs to be cleaned and transformed to correspond to the designed model

```
df = read.csv("raw_data/credit-1.csv", stringsAsFactors = TRUE)
df$Is_house_owner <- as.numeric(df$home == "owner")
df$Is_married <- as.numeric(df$marital == "married")
df$has_fixed_job <- as.numeric(df$job == "fixed")
df$is_freelance <- as.numeric(df$job == "freelance")
df$is_parttime <- as.numeric(df$job == "parttime")
df$reimbursement_cap <- df$expenses/df$income
df$income_to_total_debt <- df$income/(df$debt+df$amount)
df$savings_to_total_debt <- df$savings/(df$debt+df$amount)
df$log_income <- log(1 + df$income)
df$log_assets <- log(1 + df$assets)

df = df[,c("default", "seniority", "age", "Is_house_owner", "Is_married",
           "has_fixed_job", "is_freelance", "is_parttime", "reimbursement_cap",
           "income_to_total_debt", "savings_to_total_debt", "log_income",
           "log_assets")]
```

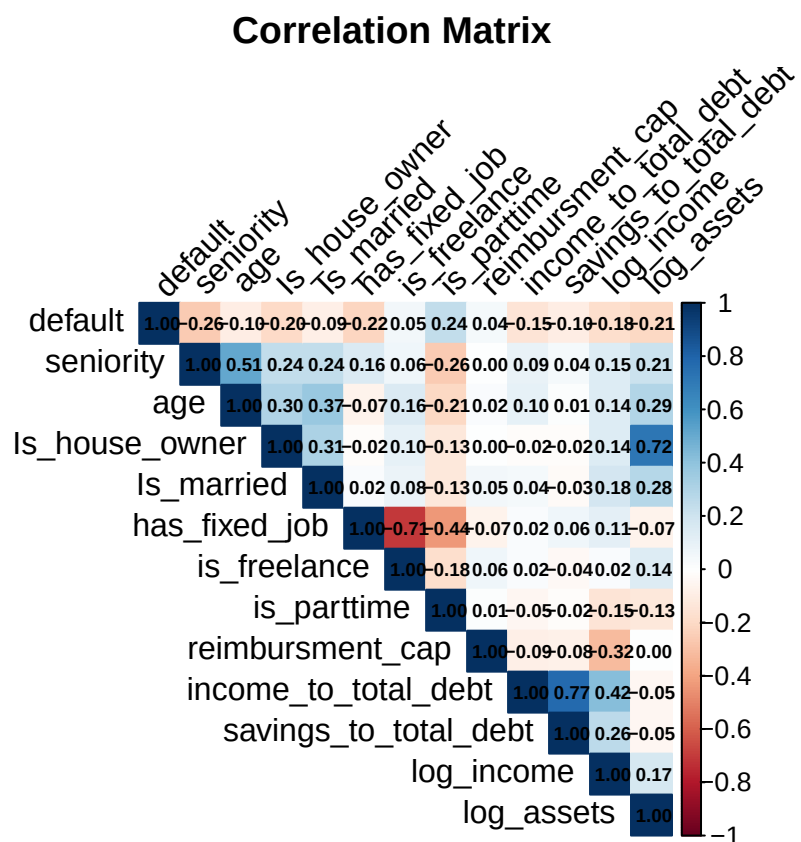
Check colinearity and data :

The colinearity between all the variables needs to be computed to check the multicollinearity hypothesis and indicate if the dependant variable has a linear relationship with the independent variables.

```
vars <- c("default", "seniority", "age", "Is_house_owner",
         "Is_married", "has_fixed_job", "is_freelance",
         "is_parttime", "reimbursement_cap",
         "income_to_total_debt", "savings_to_total_debt",
         "log_income", "log_assets")

cor_mat <- round(cor(df[, vars], use = "complete.obs"), 2)

corrplot(cor_mat,
         method = "color", type = "upper",
         addCoef.col = "black", number.cex = 0.55,
         tl.col = "black", tl.srt = 45,
         title = "Correlation Matrix", mar = c(0, 0, 2, 0))
```



Model creation

LPM model estimating

```
LPM <- lm(default ~ seniority + age + Is_house_owner + Is_married
          + has_fixed_job + is_freelance + is_parttime
          + reimbursement_cap + income_to_total_debt
          + savings_to_total_debt + log_income + log_assets, df)
```

```
summary(LPM)
```

```
##
## Call:
## lm(formula = default ~ seniority + age + Is_house_owner + Is_married +
##      has_fixed_job + is_freelance + is_parttime + reimbursment_cap +
##      income_to_total_debt + savings_to_total_debt + log_income +
##      log_assets, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.7831 -0.2948 -0.1436  0.3544  1.2686
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.6204866   0.0754642    8.222 2.60e-16 ***
## seniority      -0.0111059   0.0009330   -11.904 < 2e-16 ***
## age            0.0043582   0.0007337    5.940 3.07e-09 ***
## Is_house_owner -0.0588502   0.0179768   -3.274 0.00107 **
## Is_married      0.0080615   0.0155381    0.519 0.60391
## has_fixed_job  -0.0808852   0.0348745   -2.319 0.02042 *
## is_freelance    0.0483973   0.0355339    1.362 0.17327
## is_parttime     0.2056998   0.0387107    5.314 1.13e-07 ***
## reimbursment_cap 0.0011841   0.0055887    0.212 0.83221
## income_to_total_debt -0.5803326 0.0813647   -7.132 1.15e-12 ***
## savings_to_total_debt 2.5318810 1.0355947    2.445 0.01453 *
## log_income      -0.0439741   0.0146801   -2.995 0.00276 **
## log_assets      -0.0158250   0.0021440   -7.381 1.87e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.41 on 4433 degrees of freedom
## Multiple R-squared:  0.1701, Adjusted R-squared:  0.1679
## F-statistic: 75.73 on 12 and 4433 DF, p-value: < 2.2e-16
```

Logit model estimating

```
logit <- glm(default ~ seniority + age + Is_house_owner + Is_married
+ has_fixed_job + is_freelance + is_parttime
+ reimbursment_cap + income_to_total_debt
+ savings_to_total_debt + log_income + log_assets,
df,
family = binomial(link = "logit"))
summary(logit)
```

```
##
## Call:
## glm(formula = default ~ seniority + age + Is_house_owner + Is_married +
##      has_fixed_job + is_freelance + is_parttime + reimbursment_cap +
```

```
##      income_to_total_debt + savings_to_total_debt + log_income +
##      log_assets, family = binomial(link = "logit"), data = df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.653878   0.435399   1.502 0.133151
## seniority        -0.080514   0.006802 -11.837 < 2e-16 ***
## age              0.025749   0.004326   5.953 2.64e-09 ***
## Is_house_owner   -0.417654   0.108500  -3.849 0.000118 ***
## Is_married        0.057173   0.090183   0.634 0.526104
## has_fixed_job    -0.370176   0.189057  -1.958 0.050228 .
## is_freelance      0.353841   0.191305   1.850 0.064369 .
## is_parttime       0.905272   0.206746   4.379 1.19e-05 ***
## reimbursment_cap  0.007919   0.027655   0.286 0.774620
## income_to_total_debt -5.125830  0.667692  -7.677 1.63e-14 ***
## savings_to_total_debt 22.102766  8.043937   2.748 0.006000 **
## log_income        -0.169223   0.088445  -1.913 0.055710 .
## log_assets        -0.086903   0.012387  -7.015 2.29e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 5280.3  on 4445  degrees of freedom
## Residual deviance: 4444.4  on 4433  degrees of freedom
## AIC: 4470.4
##
## Number of Fisher Scoring iterations: 5
```

Probit estimating

```
probit <- glm(default ~ seniority + age + Is_house_owner + Is_married
+ has_fixed_job + is_freelance + is_parttime
+ reimbursment_cap + income_to_total_debt
+ savings_to_total_debt + log_income + log_assets,
df,
family = binomial( link = "probit"))

summary(probit)

##
## Call:
## glm(formula = default ~ seniority + age + Is_house_owner + Is_married +
##      has_fixed_job + is_freelance + is_parttime + reimbursment_cap +
##      income_to_total_debt + savings_to_total_debt + log_income +
##      log_assets, family = binomial(link = "probit"), data = df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.412521   0.257339   1.603 0.108929
## seniority        -0.044213   0.003739 -11.823 < 2e-16 ***
## age              0.015251   0.002536   6.014 1.81e-09 ***
```

```
## Is_house_owner      -0.231823    0.063046   -3.677 0.000236 ***
## Is_married          0.024032    0.053108    0.453 0.650904
## has_fixed_job       -0.238253    0.113453   -2.100 0.035726 *
## is_freelance        0.192282    0.115092    1.671 0.094787 .
## is_parttime         0.546775    0.124609    4.388 1.14e-05 ***
## reimbursement_cap   0.004681    0.017254    0.271 0.786159
## income_to_total_debt -2.705312    0.364961   -7.413 1.24e-13 ***
## savings_to_total_debt 11.983653    4.434062    2.703 0.006879 **
## log_income          -0.114974    0.051695   -2.224 0.026142 *
## log_assets          -0.051704    0.007296   -7.087 1.37e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 5280.3  on 4445  degrees of freedom
## Residual deviance: 4454.2  on 4433  degrees of freedom
## AIC: 4480.2
##
## Number of Fisher Scoring iterations: 5
```

Interpretation :

- *seniority* : negative coefficient as expected, significant at 99% level
- *age* : positive and significant at 99% level, meaning that the older is the client the higher is the default risk. May be due to bigger financial pressure due to children for age range 35 - 60 yo and less financial capacities for age range > 60 yo
- *Is_house_owner* : negative as expected, significant at 95% level of confidence
- *Is_married* : not significant, may be due to high financial pressure for families
- *has_fixed_job* : negative as expected, significant at 95% level of confidence
- *is_freelance* : not significant, being freelance may induce more volatility but is often related to higher incomes and more flexibility
- *is_parttime* : positive as expected, significant at 99% level of confidence *reimbursement_cap* : not significant, may be due to the fact that expenses changes when people loan money, so the reimbursement_cap ratio is not relevant anymore
- *income_to_total_debt* : negative and large meaning it has a big impact on the default capacity, which make sense because the risk of default is related to the value of the debt compared to the ability to reimburse this debt
- *savings_to_total_debt* : surprisingly positive, may be due to colinearity with *income_to_total_debt*
- *log_income* and *log_asset* : both negative and significant at 95% level of confidence

Application of the model to 10 personas

We will now use the 3 models created to estimate the probabilities of default of 10 clients that asked for a loan.

Data Preparation

```
to_predict <- read.csv("raw_data/prospects.csv",stringsAsFactors = TRUE)
summary(to_predict)
```

```
##      prospect  seniority      home      age      marital
## Beckmann   :1  Min.    : 0.0  owner   :7  Min.    :23.00  married  :7
## Buntschuh  :1  1st Qu.: 2.0  parents:1 1st Qu.:27.75  separated:1
## Fassbender:1  Median : 4.0  rent    :2  Median :32.50  single   :2
## Ginter     :1  Mean    : 6.1                Mean    :36.20
## Heckmann   :1  3rd Qu.: 9.5                3rd Qu.:45.50
## Kremer     :1  Max.    :19.0                Max.    :56.00
## (Other)    :4
##      job      expenses      income      assets
## fixed      :4  Min.    :35.00  Min.    : 101  Min.    : 0
## freelance:5  1st Qu.:37.50  1st Qu.: 150  1st Qu.: 6000
## parttime   :1  Median :60.00  Median : 286  Median : 11250
##            Mean    :58.50  Mean    :1608  Mean    : 50000
##            3rd Qu.:71.25  3rd Qu.: 395  3rd Qu.: 18375
##            Max.    :90.00  Max.    :13700  Max.    :400000
##
##      debt      amount      savings
## Min.    : 0.0  Min.    : 370.0  Min.    :-0.300
## 1st Qu.: 0.0  1st Qu.: 781.2  1st Qu.: 2.801
## Median : 0.0  Median : 950.0  Median : 4.100
## Mean    :585.0  Mean    : 944.5  Mean    : 4.068
## 3rd Qu.:462.5  3rd Qu.:1150.0  3rd Qu.: 5.287
## Max.    :4000.0  Max.    :1550.0  Max.    : 7.824
##
```

```
to_predict$Is_house_owner <- as.numeric(to_predict$home == "owner")
to_predict$Is_married <- as.numeric(to_predict$marital == "married")
to_predict$has_fixed_job <- as.numeric(to_predict$job == "fixed")
to_predict$is_freelance <- as.numeric(to_predict$job == "freelance")
to_predict$is_parttime <- as.numeric(to_predict$job == "parttime")
to_predict$reimbursement_cap <- to_predict$expenses/to_predict$income
to_predict$income_to_total_debt <-
  to_predict$income/(to_predict$debt+to_predict$amount)
to_predict$savings_to_total_debt <-
  to_predict$savings/(to_predict$debt+to_predict$amount)
to_predict$log_income <- log(1 + to_predict$income)
to_predict$log_assets <- log(1 + to_predict$assets)

to_predict = to_predict[,c("seniority","age","Is_house_owner","Is_married",
  "has_fixed_job","is_freelance","is_parttime",
  "reimbursement_cap","income_to_total_debt",
  "savings_to_total_debt","log_income",
  "log_assets","amount","prospect")]
```

Predict the probabilities for the prospect using the three different models

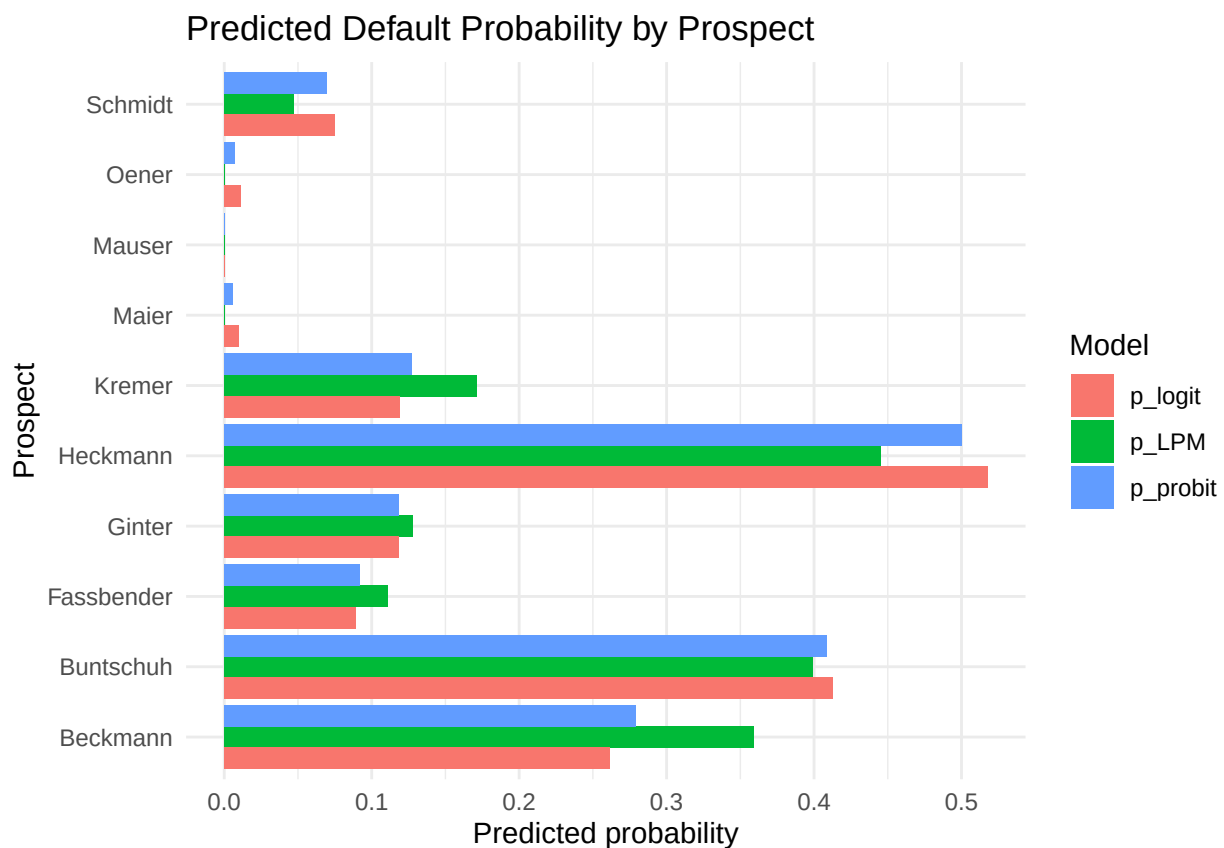
```
to_predict$p_LPM <- pmin(1,pmax(0, predict(LPM, newdata = to_predict)))
to_predict$p_logit <- predict(logit, newdata = to_predict, type = "response")
to_predict$p_probit <- predict(probit, newdata = to_predict, type = "response")
```

Models estimated probabilities comparison :

```
probabilities <- to_predict %>%
  select(prospect, p_LPM, p_logit, p_probit) %>%
  pivot_longer(
    cols = c(p_LPM, p_logit, p_probit),
    names_to = "Model",
    values_to = "Probability"
  )

probabilities_comparison_plot <- ggplot(probabilities,
  aes(x = prospect, y = Probability, fill = Model)) +
  geom_col(position = "dodge") +
  coord_flip() +
  labs(
    title = "Predicted Default Probability by Prospect",
    x = "Prospect",
    y = "Predicted probability",
    fill = "Model"
  ) +
  theme_minimal()

print(probabilities_comparison_plot)
```



```
ggsave( filename = "Output_plots/probabilities_comparison.png",
        plot = probabilities_comparison_plot ,
        width = 9,
        height = 6,
        dpi = 300 )
```

Application of the result to a 1- year loan 10% IR

We will now consider they asked for a 1-year loan with 10% interest rate, considering no cost of borrowing and no other variable costs :

Earning without default :

$$\text{Earning without default} = 0.10 * \text{Loan amount}$$

```
to_predict$earning_wo_default <- 0.10 * to_predict$amount
```

Expected earnings : *using the law of total probability*

$$\begin{aligned} \text{Expected earning} &= (1 - p_{\text{default}}) \times \text{Earning without default} \\ &\quad - p_{\text{default}} \times \text{Loan amount} \end{aligned}$$

```
to_predict$expected_earning <-
  (1 - to_predict$p_LPM) * to_predict$earning_wo_default -
  to_predict$p_LPM * to_predict$amount
```

A loan is accepted if the expected earning is positive

```
# if loan accepted : Final decision = 1
# if loan refused : Final decision = 0
to_predict$final_decision <- as.numeric(to_predict$expected_earning > 0)

print(str_glue("Number of accepted loan : {sum(to_predict$final_decision)}"))
```

```
## Number of accepted loan : 4
```

```
accepted_loan_names_list <-
  to_predict[to_predict$final_decision == 1, "prospect"]

print(str_glue("Accepted loan names list : {accepted_loan_names_list}"))
```

```
## Accepted loan names list : Maier
##Accepted loan names list : Mauser
##Accepted loan names list : Schmidt
##Accepted loan names list : Oener
```

The expected total profit would be :

```
total_expected_earning = sum(as.numeric(to_predict$final_decision) *
                             to_predict$expected_earning)

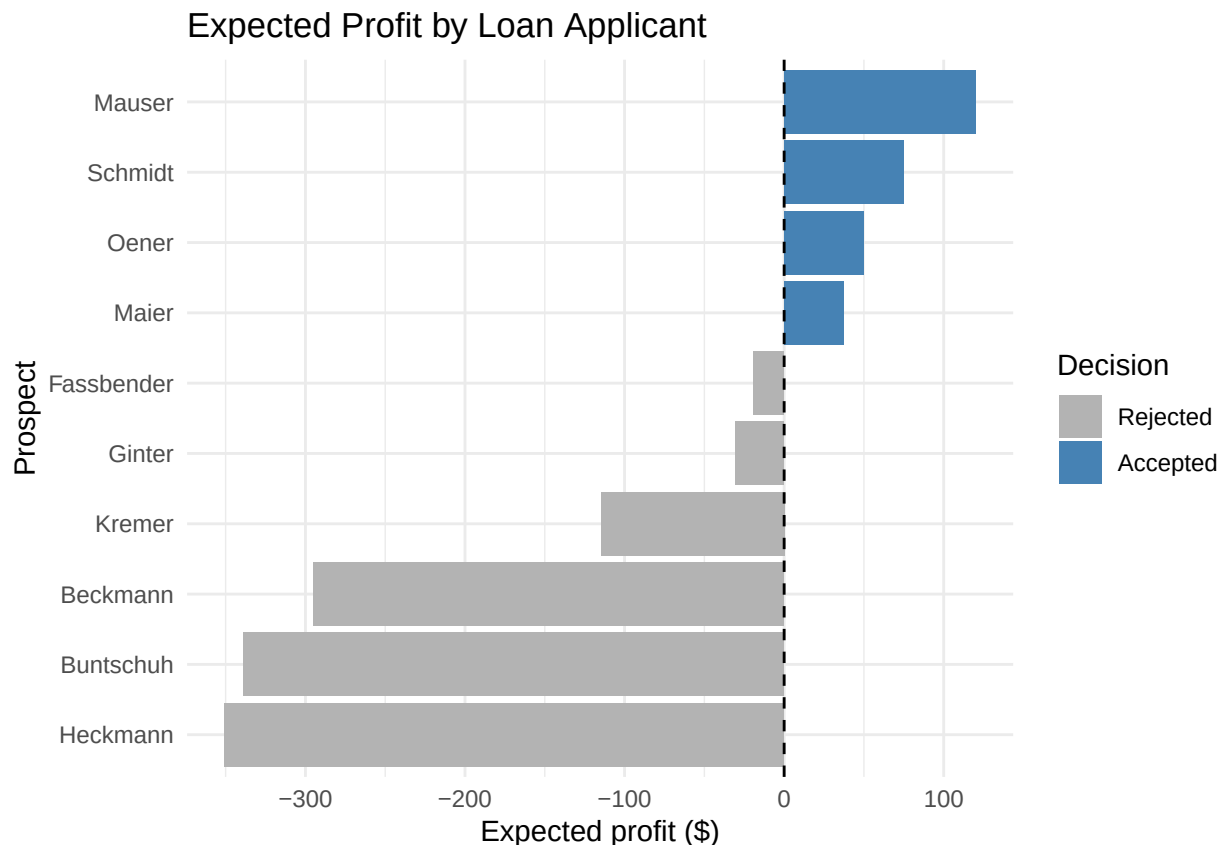
print(str_glue("expected earning : {round(total_expected_earning,2)} $"))
```

```
## expected earning : 281.97 $
```

Summary of the decision :

```
expected_profit_by_prospect <- ggplot(to_predict,
  aes(x = reorder(prospect, expected_earning),
      y = expected_earning,
      fill = factor(final_decision))) +
  geom_col() +
  coord_flip() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  scale_fill_manual(
    values = c("0" = "grey70", "1" = "steelblue"),
    labels = c("0" = "Rejected", "1" = "Accepted")
  ) +
  labs(
    title = "Expected Profit by Loan Applicant",
    x = "Prospect",
    y = "Expected profit ($)",
    fill = "Decision"
  ) +
  theme_minimal()

print(expected_profit_by_prospect)
```



```
ggsave( filename = "Output_plots/expected_profit_by_prospect.png", plot =
  expected_profit_by_prospect , width = 9, height = 6, dpi = 300 )
```

Bad Scenario

To limit the risk at the maximum we can use the upper bound of the probability interval

```
#predict probability with prediction interval
prediction_interval <- predict(LPM,newdata = to_predict,interval="prediction",
  level = 0.95)

to_predict$Bad_Scn_p_LPM <- pmin(1,pmax(0, prediction_interval[, "upr"]))

#compute expected earning
to_predict$Bad_Scn_expected_earning <-
  (1 - to_predict$Bad_Scn_p_LPM) *
  to_predict$earning_wo_default -
  to_predict$Bad_Scn_p_LPM * to_predict$amount

#compute final decision
to_predict$Bad_Scn_final_decision <-
  as.numeric(to_predict$Bad_Scn_expected_earning > 0)

print(
  str_glue("Number of accepted loan : {sum(to_predict$Bad_Scn_final_decision)}"))
```

```
## Number of accepted loan : 1
```

```
Bad_Scn_accepted_loan_names_list <-  
  to_predict[to_predict$Bad_Scn_final_decision == 1,"prospect"]  
  
#compute total expected earning  
Bad_Scn_total_expected_earning =  
  sum(as.numeric(to_predict$Bad_Scn_final_decision)  
    * to_predict$Bad_Scn_expected_earning)  
  
print(str_glue("Accepted loan names list : {Bad_Scn_accepted_loan_names_list}"))
```

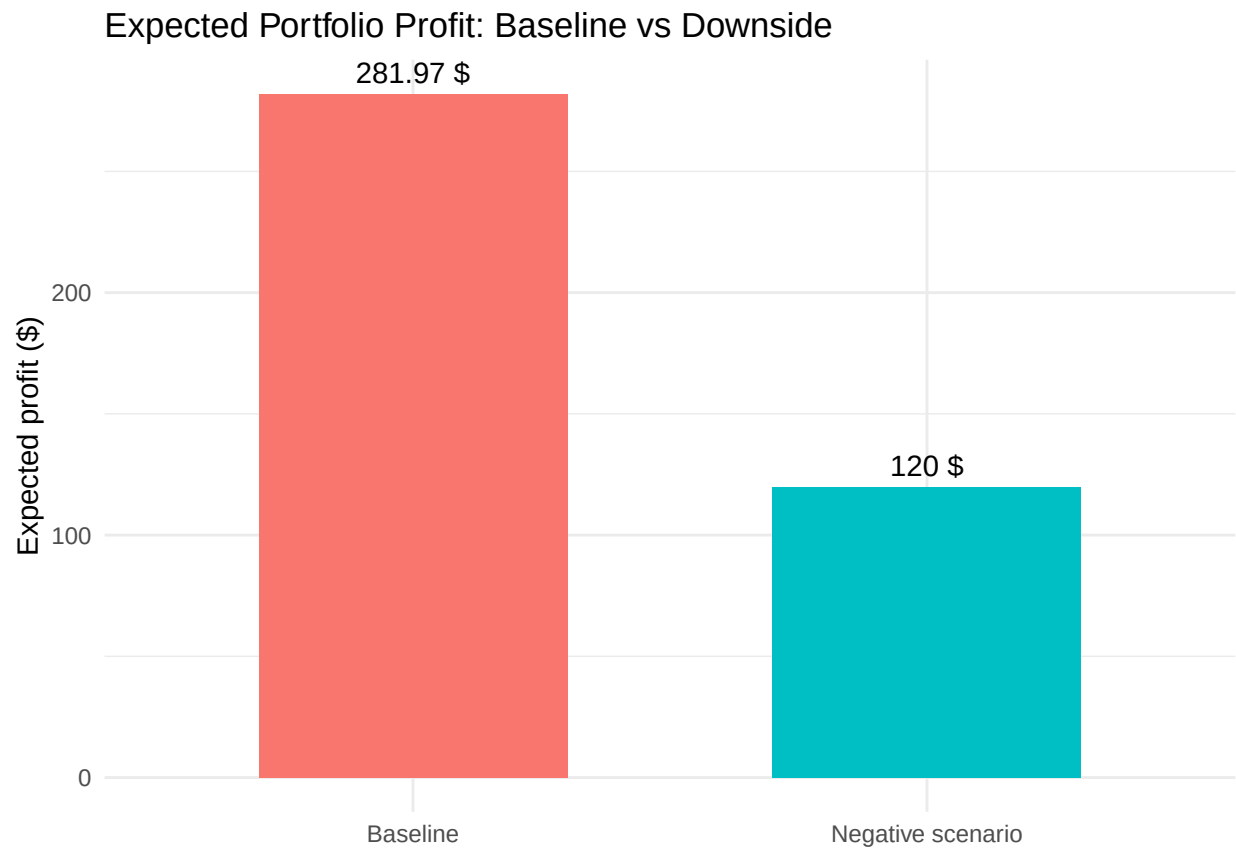
```
## Accepted loan names list : Mauser
```

```
print(str_glue("expected earning : {round(Bad_Scn_total_expected_earning,2)} $"))
```

```
## expected earning : 120 $
```

Visual comparison between bad scenario and baseline

```
scenario_results <- data.frame(  
  Scenario = c("Baseline", "Negative scenario"),  
  Expected_Profit = c(  
    total_expected_earning,  
    Bad_Scn_total_expected_earning  
  )  
)  
  
scenarii_comparison <- ggplot(scenario_results,  
  aes(x = Scenario, y = Expected_Profit, fill = Scenario)) +  
  geom_col(width = 0.6) +  
  geom_text(  
    aes(label = paste0(round(Expected_Profit, 2), " $")),  
    vjust = -0.5  
  ) +  
  labs(  
    title = "Expected Portfolio Profit: Baseline vs Downside",  
    x = NULL,  
    y = "Expected profit ($)"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")  
  
print(scenarii_comparison)
```



```
ggsave(  
  filename = "Output_plots/scenarii_comparison.png",  
  plot = scenarii_comparison ,  
  width = 9,  
  height = 6,  
  dpi = 300 )
```